SOAL 2 ENSKRIPSI CITRA LOGO TOKO DENGAN INVERS MATRIX 3x3

$a = 3 \;.\; b = 3$ $\quad A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix} \rightarrow A = \begin{bmatrix} 4 & 3 & 2 \\ 1 & 9 & 3 \\ 3 & 1 & 6 \end{bmatrix}$

a). Determinan ekspansi kofactor baris 1 :

> $a_{11} \cdot C_{11} = 4 \times (+1) \times \det[[9,3],[1,6]] = 4 \times 51 = 204$
> $a_{12} \cdot C_{12} = 3 \times (-1) \times \det[[1,3],[3,6]] = 3 \times (-1) \times (-3) = +9$
> $a_{13} \cdot C_{13} = 2 \times (+1) \times \det[[1,9],[3,1]] = 2 \times (-26) = -52$
> $\det(A) = 204 + 9 - 52 = 161$

b). 9 Minor M_{ij} :

$i = 1 \rightarrow j = 1 \rightarrow M_{11} = 51$
$j = 2 \rightarrow M_{12} = -3$
$j = 3 \rightarrow M_{13} = -26$

$i = 2 \rightarrow j = 1 \rightarrow M_{21} = 16$
$j = 2 \rightarrow M_{22} = 18$
$j = 3 \rightarrow M_{23} = -5$

$i = 3 \rightarrow j = 1 \rightarrow M_{31} = -9$
$j = 2 \rightarrow M_{32} = 10$
$j = 3 \rightarrow M_{33} = 33$

c). Kofaktor C . Adjoin , dan invers

$C = \begin{bmatrix} 51 & 3 & -26 \\ -16 & 18 & 5 \\ -9 & -10 & 33 \end{bmatrix}$. $Adj(A) = C^T = \begin{bmatrix} 51 & -16 & -9 \\ 3 & 18 & -10 \\ -26 & 5 & 33 \end{bmatrix}$ $\quad A^{-1} = \frac{1}{161} \begin{bmatrix} 51 & -16 & -9 \\ 3 & 18 & -10 \\ -26 & 5 & 33 \end{bmatrix}$

d). Verifikasi MATLAB ; –

e). $P = \begin{bmatrix} 120 & (50+N) & 200 \\ 80 & a+2 & (10+N) \\ (30+N) & 90 & 170 \end{bmatrix} \rightarrow \begin{bmatrix} 120 & 83 & 200 \\ 80 & 5 & 43 \\ 63 & 90 & 170 \end{bmatrix}$

Enskripsi $E = A \cdot P$:

$E = \begin{bmatrix} 846 & 527 & 1269 \\ 1029 & 398 & 1097 \\ 818 & 794 & 1663 \end{bmatrix}$

Deskripsi $P' = A^{-1} \cdot E$.
Jika $\det(A) = 0$
A tidak invertibel, deskripsi tidak bisa dilakukan.
(data hilang permanen)

Dipindai dengan CamScanner